Sensor Evaluation for Autonomous Docking Operations: A Literature Review

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Date: 10-07-2025

Submission: Second Attempt

Executive Summary

This research report was carried out for Rotterdam University of Applied Sciences, in collaboration with Sens2Sea, and focuses on identifying the most suitable sensor for the prototype of the project *"Autonomous Manoeuvring in the Harbor"*. The prototype is a sensor module mounted on a ship to determine the distance to the quay. This data enables the ship to control its motion mechanisms and thus manoeuvre autonomously in the harbour.

The main research question, *"What is the most suitable sensor for the prototype?"*, is addressed by comparing different types of sensors and analysing the influence of environmental factors on their performance. The sensors are further evaluated based on various criteria to identify the most appropriate option.

The evaluation shows that the XM125 radar sensor is the most suitable for the prototype. However, for further development, it is recommended to transition to the TI radar sensor, which, although more expensive, offers more advanced features and better overall performance.

Introduction

This research has been conducted for the project *"Autonomous Manoeuvring in the Harbor"*. The project was initiated by Geert Mosterdijk, co-owner of the company *Sens2Sea*. This company specializes in maritime radar and measurement systems.

Currently, the docking of ships is carried out manually by captains. This process does not always go smoothly or in the most efficient way. For large ships, this can lead to high energy costs and potential damage claims. For this reason, *Sens2Sea* aims to automate the docking process using sensor technology. These sensors are placed on the ship and must be able to determine the distance to the quay. Based on this data, the ship's motion mechanisms can be controlled, enabling it to manoeuvre autonomously in the harbour.

To achieve this, a prototype is first being developed in which these sensors can operate in a modular way. The goal of this research is to identify the most suitable sensor for the prototype.

To answer the main research question, *"What is the most suitable sensor for the prototype?"*, the following sub-questions are explored:

1. How do the potential distance sensors for this project differ from one another?
2. How do environmental factors affect the performance of these sensors?

Theoretical Framework

Prior to this literature review, a preliminary study was conducted to examine how the beam diameter of a sensor affects its measurements (see appendix). This study concluded that a smaller beam diameter is more suitable for the intended application, as it allows for clearer distinction between different objects.

Methods

This report answers the main and sub-research questions based on a literature review.

Research steps

1. Different Sensors

To answer the first sub-question, research is conducted into how various distance sensors work and how they differ from each other. The selected sensors must be able to measure a distance of at least 1 centimetre up to 10 meters.

2. Influence of Environmental Factors

To answer the second sub-question, research is carried out on the influence of environmental factors on the sensors. This includes examining how the sensors perform in various weather conditions and in open-air environments.

3. Sensor Evaluation

To answer the main research question, the different types of sensors are compared in a table based on their resistance to environmental factors, beam diameter size, and price. Greater resistance to environmental conditions, a smaller beam diameter, and a lower price result in a higher score.

- Resistance to environmental factors has the highest and therefore strongest weight: a weight of 6. This is because the sensor will ultimately be mounted on a ship and must function well in varying weather conditions, making this factor particularly important.

- Beam diameter size has a weight of 4. This is important because a smaller beam diameter leads to a more precise sensor, which can better distinguish between detected objects.

- Price has the lowest weight, a weight of 2. This is because cost is less important at this stage of the research. The goal is to choose the most suitable type of sensor for the project.

4. Specification of Sensor Choice

To fully answer the main research question, specific sensor models are selected from the chosen sensor type. These sensor models are then compared in a table based on their range, accuracy, price, and beam diameter. A greater range, higher accuracy, lower price, and smaller beam diameter result in a higher score.

- Price has the highest weight, with a value of 7. This is because it plays a key role in selecting the most suitable sensor for the prototype. The sensor must fit within the project's budget, which is €500.

- Accuracy has a weight of 5. This factor is important as it ensures accurate data in the prototype.

- Beam diameter also has a weight of 5. This is important because a smaller beam diameter results in a more precise sensor that can better distinguish between detected objects.

- Range has a weight of 4. This is the least important factor compared to the others, as it does not significantly affect the functioning of the prototype. A sensor with a shorter range is still sufficient for this application.

Results

This report will present the findings related to the main research question and sub-questions mentioned in the introduction.

How do the potential distance sensors for this project differ from one another?

The radar sensor performs well under various weather conditions. The longer wavelength of radio waves causes minimal signal disruption from small water droplets or dust particles. In contrast, the LiDAR sensor is sensitive to water and dust particles due to the shorter wavelength and reflectivity of light waves. Furthermore, the sonar sensor is sensitive to temperature fluctuations in the air and operates less efficiently in air than in water because air has a lower density than water, resulting in poorer sound wave propagation.

Based on the requirements previously outlined in the methodology, it appears that the radar sensor, sonar sensor, and lidar sensor are potentially suitable for the intended application of this project.

- **A sonar sensor** uses sound waves to determine distance. The sensor emits sound waves through a transducer and detects the reflected waves to calculate the distance. Sonar technology works particularly well underwater because sound waves travel efficiently in water (MindKosh®, sd).
- **A radar sensor** uses radio waves to determine distance. The sensor emits these radio waves through an antenna and detects the reflections. By measuring the time between transmission and detection of the reflection, the distance can be calculated (Electronics, 2024).
- **A lidar sensor** is based on laser technology that uses infrared or ultraviolet light. The laser beam reflects off an object and returns to the sensor. By measuring the time between emission and detection of the reflection, the distance can be calculated (Analytics, n.d.).

All three sensor types can measure distance but differ, among other things, in accuracy. The LiDAR sensor has the smallest beam diameter and therefore the highest resolution. The beam diameter of a radar sensor can be reduced by using a higher frequency, higher antenna gain, a larger antenna, or by employing a lens. These factors decrease the beam diameter and improve the sensor's accuracy. Similarly, a sonar sensor's beam diameter can be reduced by increasing the size of the transducer, raising the frequency, or using a lens. Additionally, the sensors differ in price, with the

LiDAR sensor generally being more expensive due to its high accuracy and complex data processing.

How do environmental factors affect the performance of these sensors?

The radar sensor performs well under various weather conditions. The longer wavelength of radio waves causes minimal signal disruption from small water droplets or dust particles. In contrast, the LiDAR sensor is sensitive to water and dust particles due to the shorter wavelength and reflectivity of light waves. Furthermore, the sonar sensor is sensitive to temperature fluctuations in the air and operates less efficiently in air than in water because air has a lower density than water, resulting in poorer sound wave propagation (MindKosh®, sd).

Sensor evaluation

Below is a table comparing and weighting the different sensors based on their resistance to environmental factors, beam diameter size, and price.

The radar sensor receives the highest score for resistance to environmental factors. This is because, unlike the LiDAR sensor, the radar sensor experiences minimal disruption from water and airborne particles. The sonar sensor scores lower in this category due to its reduced effectiveness in air.

The LiDAR sensor scores highest for beam diameter size, as it has the smallest beam diameter. While the beam diameters of the sonar and radar sensors can be improved, they cannot match the precision of the LiDAR sensor.

In terms of price, the LiDAR sensor receives the lowest score because its high accuracy and complex data processing generally result in a higher cost.

Sensor		**Radar**		**Sonar**		**Lidar**	
Criteria:	Weight:	Factor:	score:	Facor:	score:	Factor:	score:
Invironmental factors	6	6	36	3	18	3	18
Beam diameter	4	2	8	2	8	4	16
Price	2	2	4	2	4	1	2
Total:		Sum:	**48**	Sum:	**30**	Sum:	**36**

The radar sensor has the highest total score of all the sensors.

Specification of sensor choice

To answer the main research question, *“What is the most suitable sensor for this prototype?”*, an analysis of available radar sensors was conducted.

The sensors examined include the RCWL-051 (MakerHero, sd), a Texas Instruments variant (Instruments, sd), and the XM125 radar sensor (Datasheet). These radar sensors differ in price and quality. In the table below, the sensors are compared and weighted based on their range, accuracy, price, and beam diameter. The XM125 radar sensor achieves the highest total score.

Radar sensor type		**RCWL-051**		**Variant TI**		**XM125**	
Criteria:	Weight:	Factor:	score:	Factor:	score:	Factor:	score:
Range	4	1	4	4	16	2	8
Accuracy	5	1	5	5	25	3	15
Price	7	7	49	2	14	7	49
Beam diameter	5	1	5	5	25	2	10
Total:		Sum:	**63**	Sum:	**80**	Sum:	**82**

Conclusions/Recommendations

The goal of this research was to identify the most suitable sensor for the prototype of the “Autonomous Docking in the Harbor” project. This was achieved by analysing the differences between sonar, LiDAR, and radar sensors, and by examining how environmental factors affect their performance.
Based on the literature review and comparative analysis, the XM125 radar sensor was found to be the most suitable option for the prototype. This is primarily due to its strong resistance to environmental conditions.
It is recommended to use the XM125 radar sensor for the current prototype. However, for further development, it is advised to switch to a TI variant radar sensor, as it performs better across all criteria except price. If price becomes a less significant factor in later stages of development, the TI variant will score higher overall and therefore be the better choice. For now, the XM125 is recommended.

Glossary

Beam diameter – The beam diameter refers to the width of the beam a sensor uses to detect objects.

Transducer – A component in a sonar sensor that emits sound waves and receives the echoes.

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Appendix

Hoe beïnvloed de spreiding van de bundel van de sensor de metingen

Wat is een bundel

De bundel is een algemene term die wordt gebruikt voor de uitgezonden en/of ontvangen staling die wordt gebruikt om een meting te doen. Dit kan een lichtbundel, infrarood of zelfs een röntgenbundel zijn. Sensoren die dit gebruiken, gebruiken het om objecten en afstanden te meten. Door onderbrekingen en verandering waar te nemen.

Wat is een bundeldiameter

De bundeldiameter (ook wel spotgroote genoemd) is de breedte van de straal die een sensor gebruikt om objecten te detecteren. Een kleinere bundeldiameter maakt makkelijker/mogelijk om individuele objecten te onderscheiden, terwijl een grotere bundeldiameter ervoor kan zorgen dat meerdere objecten als één geheel worden waargenomen.

Er komen hier nog een paar belangrijke termen bij kijken:

- Divergentie(beam divergence) Hoeveel de bundel zich verspreid over een afstand, dit is belangrijk om er voor te zorgen dat meerdere objecten niet als één geheel worden gezien.
- Focusafstand(focal distance) Dit is de afstand waarop de bundel het 'scherpst' is
- Detectieresolutie(Detection resolution) Hoe nauwkeurig de sensor objecten van elkaar kan onderscheiden

Hoe beïnvloed de bundeldiameter de metingen

Er zijn meerdere manieren waarop de bundel/ bundeldiameter de metingen beïnvloed, waar rekening mee moet worden gehouden voor de uiteindelijke sensoren op be boot. Aangezien de termen grote- en kleine bundel worden gebruikt, wordt er met kleine bundel 1-5mm bedoeld en wordt er met een grote bundel 10-50mm bedoeld.

1. Resolutie van de meting:

- Kleine bundel: Een kleinere bundel kan nauwkeurigere metingen opleveren doordat deze een kleiner gebied bestrijkt. Dit is vooral belangrijk in toepassingen waar precisie vereist is, zoals bij optische sensoren voor objectherkenning.

- Grote bundel: Een grotere bundel kan minder gedetailleerde informatie geven omdat het een groter gebied bestrijkt en dus mogelijk meer ruis of storingen van omgevingsfactoren opvangt.

2. Scherpte van de meting:

- Kleine bundel: Het kan moeilijk zijn om kleine, goed afgelijnde objecten te detecteren, omdat de bundel heel precies moet zijn om een object te raken. In sommige gevallen kan het ook moeilijker zijn om een goed signaal te krijgen als de bundel te smal is.

- Grote bundel: Een grotere bundel is mogelijk makkelijker in staat om objecten te detecteren, zelfs als ze zich niet precies in het midden van de bundel bevinden, maar het kan ook minder scherp zijn bij het onderscheiden van objecten of veranderingen in het gemeten gebied.

3. Signaalsterkte:

- Kleine bundel: Het kan resulteren in een hogere intensiteit van het signaal op een klein gebied, wat handig is voor het detecteren van kleine hoeveelheden licht of specifieke details. Maar het kan ook gevoelig zijn voor kleine afwijkingen of verstoringen.
- Grote bundel: Een bredere bundel kan het signaal over een groter gebied verspreiden, waardoor het gemiddelde signaal zwakker is, maar tegelijkertijd minder gevoelig voor kleine verstoringen.

4. Meetbereik:

- Kleine bundel: Een kleinere bundel kan de meetcapaciteit beperken tot een specifiek, beperkt bereik. Dit is handig voor gedetailleerde, lokale metingen, maar minder geschikt voor het detecteren van bredere gebieden.
- Grote bundel: Een bredere bundel kan een groter gebied bestrijken, wat nuttig is voor het meten of detecteren van objecten over een groter bereik, maar dit kan ten koste gaan van de precisie.

5. Interferentie en ruis:

- Kleine bundel: De kans op interferentie kan groter zijn omdat kleine afwijkingen in de sensor of omgeving een grotere invloed hebben op de metingen.

- Grote bundel: Een bredere bundel kan minder gevoelig zijn voor kleine storingen, omdat het signaal over een groter gebied wordt verspreid, maar kan ook meer achtergrondruis oppikken van het grotere meetgebied.

6. Bewegingsgevoeligheid:

- Kleine bundel: Bij snel bewegende objecten kan de kleine bundel moeite hebben met het volgen van de beweging, omdat het sneller moet reageren op veranderingen in positie.

- Grote bundel: Een bredere bundel kan gemakkelijker meebewegen met objecten die zich over een groter gebied bewegen, maar kan tegelijkertijd minder nauwkeurig zijn in het volgen van specifieke bewegingen.

Conclusie

Na het kijken, naar alle mogelijke manieren waarop de bundeldiameter de metingen beïnvloed en die 'tegen de eisen aan te leggen'. Is de keuze gemaakt om een zo klein mogelijke bundeldiameter te hanteren, waar mogelijk is.